

Problem 6.16

Carl puts 10,000 into a bank account that pays an annual effective interest rate of 4% for ten years. If a withdrawal is made during the first five and one-half years, a penalty of 5% of the withdrawal amount is made. Carl withdraws K at the end of each of years 4, 5, 6, 7. The balance in the account at the end of year 10 is 10,000. Calculate K .

The initial deposit grows to

$$10000(1.04)^{10}$$

by the end of year 10.

The withdrawals at the end of years 4 and 5 are within the first 5.5 years, so each incurs a 5% penalty. Thus each of these reduces the account by $1.05K$.

The withdrawals at the end of years 6 and 7 do not incur a penalty, so each reduces the account by K .

Accumulating all values to the end of year 10, we get

$$10000(1.04)^{10} - 1.05K(1.04)^6 - 1.05K(1.04)^5 - K(1.04)^4 - K(1.04)^3 = 10000.$$

Using

$$10000(1.04)^{10} \approx 14802.4428,$$

this becomes

$$14802.44 - K [1.05(1.04)^6 + 1.05(1.04)^5 + (1.04)^4 + (1.04)^3] = 10000.$$

So,

$$14802.4428 - 4.80256K = 10000.$$

Hence,

$$4.90079K = 4802.4428,$$

and therefore

$$K \approx 979.931763$$